

1 Counterfactual crisis generation

Isomorph-Reborn

NETWORK

Any multi-echelon network, here ISOMORPH (13 sites, 16 links). Two representations: instantaneous capacitated flow, or travel lead times with (s, S) inventories.

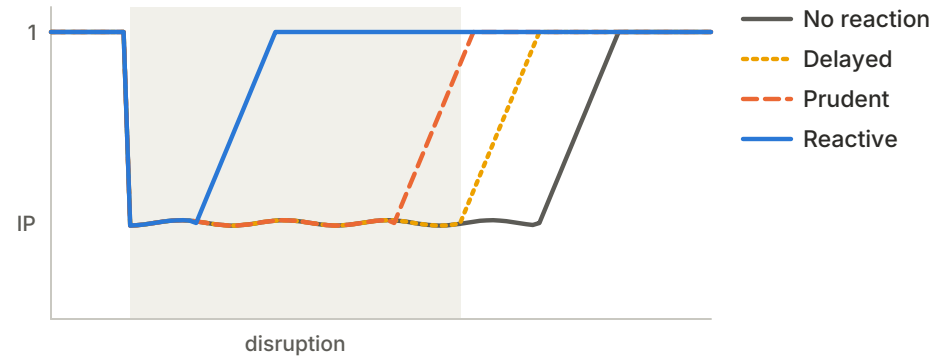
CONTROLLED DISRUPTION

Five types: supplier failure, link cut, storage site closure, demand surge, congestion. Target, severity and duration are known.

SIMULATED OPERATIONS MANAGER

Smoothed service level, detection threshold, decision delay, reassessment, then escalation among 9 levers: rerouting, capacity, inventory, demand.

SAME CRISIS, FOUR VERSIONS



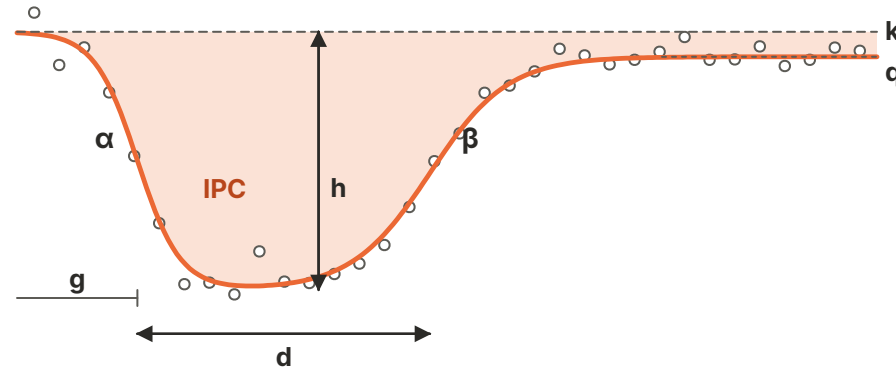
OUTPUT

8,000 daily service trajectories $IP(t)$: 2,000 crises, each played without reaction and under three profiles, with the decision log.

2 Resilience characterization

curve fitting

FITTED CURVE AND PARAMETER READING



MODEL

Composite hyperbolic curve with seven parameters, fitted with a robust loss from multiple starting points.

SIX ANALYTICAL INDICATORS

IPC (cumulative loss, prognosis target), TRP (recovery time), CRD (dynamic recovery), SRA, ERR, PR.

INTERPRETATION

h : shock depth; d : duration of the degraded plateau; α , β : abruptness of the drop and speed of recovery; g : absorption; q : lasting loss.

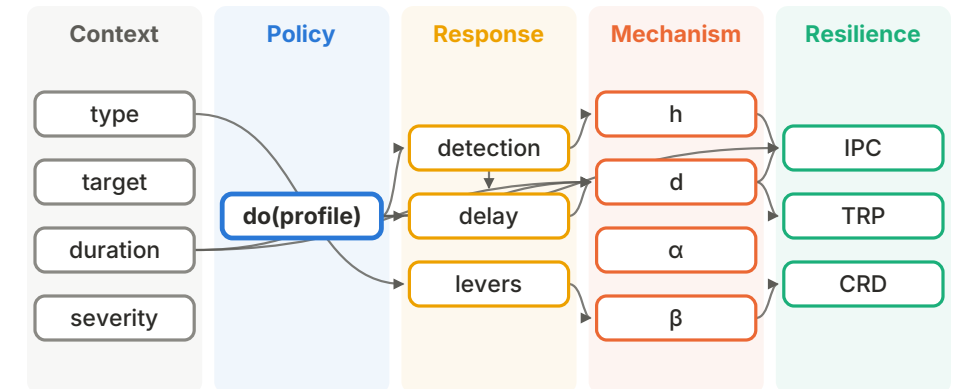
OUTPUT

Learning dataset of 20 variables: context, policy, response, mechanism (h , d , α , β) and resilience (IPC, TRP, CRD), discretized into equal-frequency classes.

3 Causal learning and decision support

BayesArtist

CAUSAL STRUCTURE CONSTRAINED BY TEMPORAL ORDER



LEARNING

Hill-climbing structure search with the BIC score, 155 forbidden arcs (no arc goes back in time), at most 4 parents; Laplace-smoothed probabilities.

PROGNOSIS

$P(\text{loss} \mid \text{context}, \text{do}(\text{policy}))$ for each candidate policy.

DIAGNOSIS

$P(\text{causes} \mid \text{high loss})$: shock, chokepoint, perception, inertia, levers.

RECOMMENDATION

The leanest policy whose probability of meeting the target exceeds a threshold θ .

VALIDATION 1,600 crises are used for learning and 400 new crises for testing. Since each test crisis was played under all four policies, every recommendation is checked against what would actually have happened.